

GLOBAL DIGITAL SERVICES



Coforge

1 Billion dollar IT services company

*Asish Kumar Khandual(85009827)
Graduate Engineer Trainee*



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What ?

Testing is a process of execution of a program ,
Intention is to find a bug. The main importance of testing is to meet the quality with respect to client req.

Why ?

Testing is needed to identify defects early, ensure quality and reliability, and reduce the risk of failures that can impact users and business.

When ?

Testing is required throughout the SDLC—starting from requirement analysis and continuing through development, deployment, and maintenance.

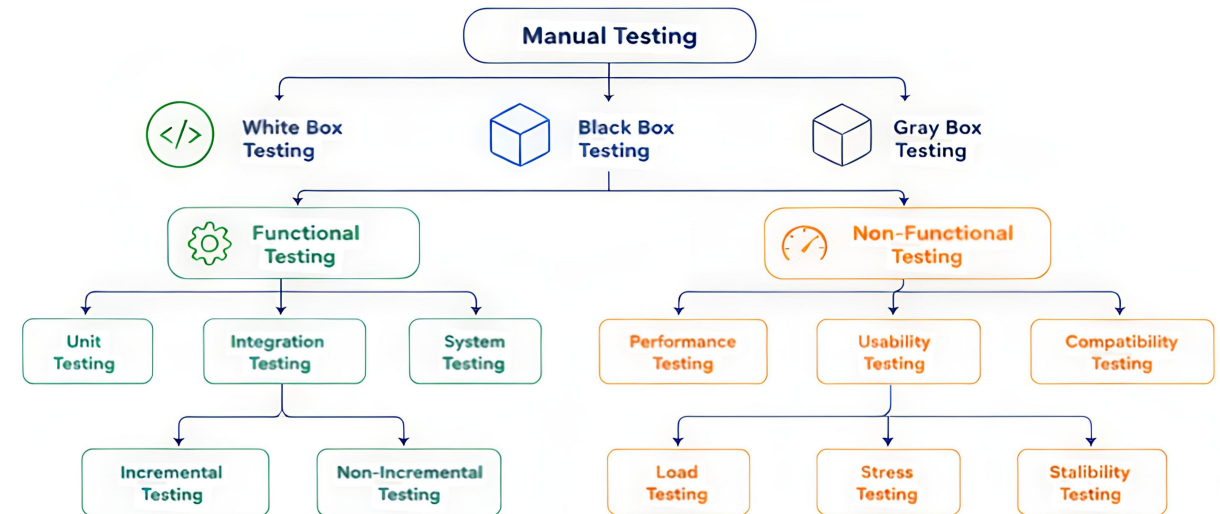


Manual Testing

Manual testing is the process of testing software without using automation tools.

Testers execute test cases **manually** to find defects.

Manual testing is needed to understand the application from a **user's perspective** , to identify **usability and visual issues** that automation cannot detect



Life Cycle Of Manual Testing



Test Plan

Test Plan for HDFC.BANK.com Website

1. Test Plan Identifier

File Name: 1.0_HDFC_tplan_v1.0.doc

- Document Name: Test Plan – HDFC Banking Application
- Version: HDFC_BA_1.0
- Date: 27-may-2026
- Location: Coforge Ltd, Hyderabad

2. Document History

REV	Author	Edits	Date
1.0	Asish Kumar Khandual, Graduate Engineer Trainee	Initial Draft	27/05/2026

3. Introduction

HDFC Bank Ltd. (HDFC) is a major Indian financial institution providing a full spectrum of banking services to retail, MSME, agricultural, and corporate clients. Originally established as a development financial institution, it has evolved into a universal bank. The bank is currently categorized by the RBI as a **Private Sector Bank** for regulatory purposes, though it remains under the joint ownership of the Life Insurance Corporation of India (LIC) and the Government of India.

4. Test Items

Specifies **what components/modules** will be tested.

Example:

- Customer Registration module
- Customer LoginModule
- Internet Banking Dashboard Module
- Account Summary Module
- Fund Transfer Module
- Beneficiary Management Module
- Bill Payment / Recharge Module
- Loan Application Module
- Logout Functionality
- Transaction History / Mini Statement Module
- Logout Functionality

5. Features to be Tested

Medium priority features are features that have been stable and High Priority features are features that have been recently changed.

Feature to be Tested	Priority	Comments
User Registration	High	
Secure Login(OTP/2FA)	High	
Dashboard Access	High	
Balance Inquiry	High	
Fund Transfer	High	
Transaction History	High	
Beneficiary Management	High	
Profile Update	High	
Logout	High	

6. Features NOT to be Tested

- Help Contents and user manuals will not be tested.
- Installation and presentation will not be tested.
- Support of the other architectures also will not be tested.
- Third-party payment gateway internal logic
- Legacy modules not impacted by current release
- Experimental features

7. Test Environment

Device specifications		Copy	^
Device name	LP-AZ-3NEL/A9DI		
Full device name	LP-AZ-3NEL/A9DI-IN-COFORGETECH.COM		
Processor	11th Gen Intel(R) Core(TM) i5-1145G7 @ 2.60GHz (1.50 GHz)		
Installed RAM	16.0 GB (15.7 GB usable)		
Device ID	85BA0706-222D-4B83-B56D-25FEA7DEBACA		
Product ID	00329-00000-00003-AA652		
System type	64-bit operating system, x64-based processor		
Pen and touch	No pen or touch input is available for this display		
Related links		Domain or workgroup	System protection
		Advanced system settings	
Windows specifications		Copy	^
Edition	Windows 11 Enterprise		
Version	23H2		
Installed on	27-03-2026		
OS build	26200.8390		
Experience	Windows Feature Experience Pack 1000.26100.297.0		
Microsoft Services Agreement			

8. Test Approach / Test Strategy

Describes **how testing will be performed.**

The test strategy for HDFC1.0 will include manual testing with automated testing(Selenium With Java) where time/need permits. Automated testing will be used to test the basic Interface objects keyboard and mouse navigation or any redundant tasks best translated to automation. Automated testing will be limited to a Win-64 platform because of Automation tool limitations.

- Sanity test will include an very short battery of tests to ensure the HDFC1.0 runs/launches and can connect to the back end database(MySql /SQL Server 2025). Test execution will happen as individual component testing. These tests will be fully automated.
- Acceptance tests will include a short battery of tests that will only touch major features to ensure HDFC1.0 will operate smoothly at a basic level. These tests will be executed on **Windows 11 Enterprise**. The goal will be to automate all of these tests.
- Full Functional tests will include a **comprehensive test** of all features listed in this document. These tests will be limited to the Windows 11 Enterprise. These test will include automation with time allowing.

8.1 Types of Testing

- Smoke Testing
- Sanity Testing
- Functional Testing
- Integration Testing
- System Testing
- Regression Testing

8.2 Test Levels

- Unit Testing (by developers)
- Integration Testing
- System Testing
- Acceptance Testing

8.3 Test Techniques

- Black Box Testing
- Boundary Value Analysis
- Equivalence Partitioning
- Error Guessing **Functionality Testing**

Usability Testing

The usability testing will be accomplished by verifying the information in each window are accurate. Menus, icons and toolbar functionality will be tested as applicable to the navigation and results panes. Multi Window Overlapping will be tested because product supports opening of multiple documents.

Robustness/Reliability Testing

Test the GUI for correct keyboard and mouse navigation, windows, menus, buttons, etc. These tests will include Keyboard only navigation of HDFC1.0, Mouse only navigation. Correctness and existence of warning/error dialog boxes. The correct/expected functionality of buttons, icons and menus.

Stress Testing
Not applicable

9. Entry and Exit Criteria

9.1 Entry Criteria

Conditions to start testing:

- Requirements approved
- Test environment ready
- Test data available
- Build deployed

9.2 Exit Criteria

Conditions to stop testing:

- All planned test cases executed
- Critical and high defects closed
- Test summary report prepared

10. Test Deliverables

Test plan	1.0_HDFC_tplan.doc
Test Specification	1.0_HDFC_tspec.doc
Test Automation Plan	TBD()
Test Checklist	TBD
Test Logs	TBD
Test tools	Selenium 4.38, Postman, Rest Assured, Jmeter, Jenkins, Git and Git

Hub and Appium and MySql Server 2025

11. Testing Tasks & Resource Needs

- Use of MySql Server database.
 - Write Test Specification.
 - Write Test Automation plan.
 - Write Test Checklist.
 - Setup workspace.
 - Develop Automation script.
- Minimum requirements for platforms:

12. Schedule

Test Documentation

Document	Development Time		Sign-off Date	Dependencies
	Person-Days	Completion Date		
Test Plan	1 day	27/05/2026	27/05/2026	Functional Spec. Review
Test Specifications	1 day	27/05/2026	27/05/2026	Test plan, Functional Spec, Review
Test Logs	TBD	27/05/2026		
Test Summary Report	TBD	27/05/2026		
Checklist	TBD	27/05/2026		

Test Development

Deliverable	Development Time		Dependencies
	Person-Days	Completion Date	
Functional Test Suite	1	27/05/2026	
Automation build			

Test Execution

Test Activity		Execution Time (Person-Days)			Dependencies
		Alpha	Beta	FCS	
Acceptance		1 Day	1 Day	1Days	
Functional	Functionality	1	1		
	Usability	All High to Medium test cases			
	Robustness				
	Stress				
	Multi Inst.				
Regression					
Documentation Review		1	1	1	
Total Test Cycle		1	1	1	

13. Risks and Assumptions

Risks:

- **Environment Instability:** Delays in setting up the test environment could impact the schedule- VM is not working port no.-5103
- **Resource Availability:** Limited availability of testers could delay test execution.
- **Late Requirement Changes:** Changes in requirements during the testing phase could lead to additional testing efforts.
- **Contingencies:**
- **Backup Resources:** Have backup resources ready to join the testing effort if required.
- **Flexible Schedule:** Allocate buffer time in the schedule to accommodate unexpected delays.
- **Regular Communication:** Hold daily stand-up meetings to quickly address any issues that arise

14. Approvals \Sign of

POSITION	NAME	SIGNATURE	REVIEWED	APPROVAL
*Approval required Project Manager				
Project Manager			Raghavendra BN	Raghavendra BN
Client Approval	Coforge			

Test Scenarios

Name: Asish Kumar Khandual
Email ID: kumarkhandual@gmail.com
Emp ID: 85009827

TEST SCENARIOS FOR HDFCBANK.COM

Scenario ID	Scenario Name	Scenario Description	Preconditions	Expected Outcome
TS_01	Verify user registration	Validate that a new user can register using valid details	User is not registered	User account is created successfully
TS_02	Verify user login	Validate that a registered user can log in with valid credentials	User must be registered	User is logged in and redirected to dashboard
TS_03	Verify successful dashboard access	Validate that user is redirected to dashboard after successful login	User is logged in	User dashboard is displayed successfully
TS_04	Verify balance inquiry	Validate that user can view account balance	User is logged in	Correct account balance is displayed
TS_05	Verify fund transfer	Validate that user can transfer funds to another account	User is logged in and beneficiary added	Funds transferred successfully
TS_06	Verify transaction history	Validate that user can view past transactions	User is logged in	Transaction history is displayed correctly
TS_07	Verify beneficiary management	Validate that user can add/edit/delete beneficiaries	User is logged in	Beneficiary is managed successfully
TS_08	Verify profile update	Validate that user can update personal details	User is logged in	Profile details updated successfully
TS_09	Verify logout functionality	Validate that user can logout securely	User is logged in	User is logged out and redirected to login page

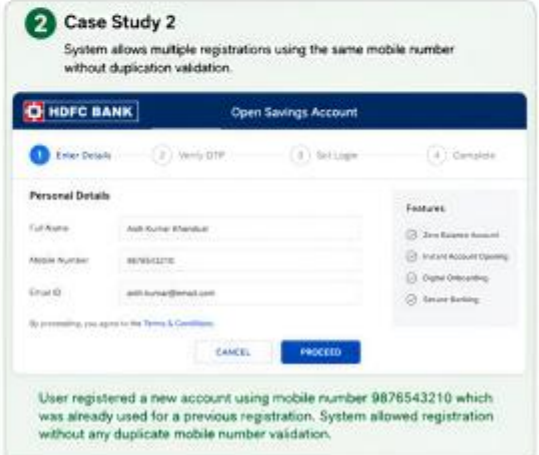
Test Cases

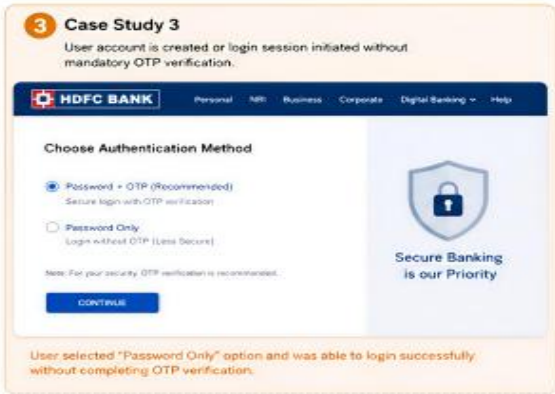
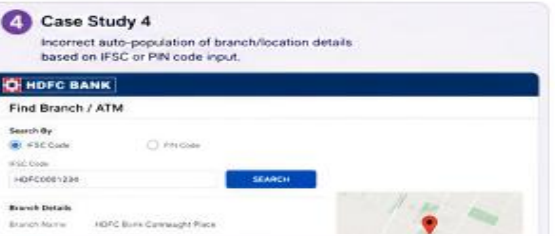
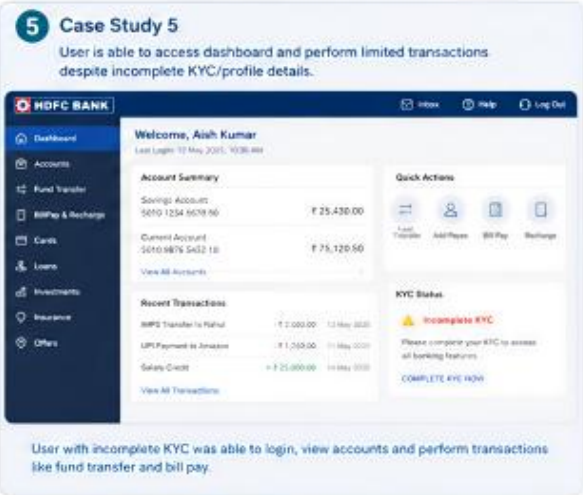
A	B	C	D	E	F	G	H
Project Name:	hdfc.bank.com						
Module Name:	End to End Testing	Date Updated:	28-05-2026				
Last Build Tested:		Reviewed By:	Raghavendra BN				
Run No:	9 Test Cases	Executed By:	Asish KumarKhandual				
Run Date:	28-05-2026						
Test Objectives:	Mapping Test Scenarios with Test Cases (hdfc.bank.in)						
Test Setup:	MS Excel Sheet (Online)						
Req_Id	Test_Case ID	Test_Case_Description	Test_steps	Test_Data	Expected_Result	Actual_Result	Status
Req_1.0	TC_01	Verify user can register with valid details	1.Launch Microsoft Edge browser on the system 2.Enter the HDFC Bank official website URL in the address bar 3.Navigate to the “Register / New User” section on homepage 4.Enter valid customer details such as Full Name, Email ID, Mobile Number. 5.User should create password as per system requirements 6.User should confirm password in confirm field 7.User should fill location details such as city/state 8.User should accept terms and conditions checkbox 9.User should click on “Register” button. 10.User should enter OTP received on registered mobile number 11.User should submit OTP to complete registration process	Url: https://www.hdfc.bank.in/ Name: Asish Kumar Email: asish@test.com Mobile: 9876543210 Password: ***** City: Sonapat	User should be successfully registered and redirected to account/dashboard setup	User registered successfully and redirected to OTP verification	Passed

Req_1.1	TC_02	Verify user can login with valid credentials	1.User should open Microsoft Edge browser 2. User should navigate to https://www.hdfc.bank.in/ 3.User should click on "Login" option 4.User should enter Customer ID or registered email 5.User should enter password 6.User should click on "Login" button 7.User should complete OTP/authentication step if prompted 8.User should land on dashboard after successful login	Url: https://www.hdfc.bank.in/ Email: asish@test.com Password: *****	User should be logged in successfully and Dashboard page should be displayed	User logged in successfully and Dashboard page should be displayed	Passed
Req_1.2	TC_03	Verify that dashboard is displayed with correct details after login	1.User should open Microsoft Edge browser 2. User should navigate to https://www.hdfc.bank.in/ 3.User should login into HDFC application 4.User should view account summary on dashboard 5.User should check available balance and account details 6.User should scroll through dashboard sections 7.User should click on different menu options (Accounts, Cards, Transfers) 8.User should verify all dashboard elements are accessible	Url: https://www.hdfc.bank.in/ Email: asish@test.com Password: *****	User should be redirected to Dashboard page and all dashboard elements should be displayed correctly	User redirected to Dashboard page and all dashboard elements should be displayed correctly	Passed
Req_1.3	TC_04	Verify user can view account balance after login	1.User should open Microsoft Edge browser 2. User should navigate to https://www.hdfc.bank.in/ 3.User should enter valid login credentials 4.User should click on "Login" button 5.User should be redirected to dashboard page 6.User should locate "Account Summary" or "Balance" section 7.User should click on account details link 8.User should verify displayed account balance	Url: https://www.hdfc.bank.in/ User ID: valid_user Password: *****	Correct account balance should be displayed	Correct account balance displayed	Passed
Req_1.4	TC_05	Verify user can transfer funds to another account	1.User should open Microsoft Edge browser 2. User should navigate to https://www.hdfc.bank.in/ 3.User should login with valid credentials 4.User should navigate to "Funds Transfer" section 5.User should select beneficiary from list 6.User should enter transfer amount 7.User should select transfer type (NEFT/IMPS/RTGS) 8.User should click on "Transfer" button 9.User should confirm transaction using OTP 10.User should verify transaction success message	Url: https://www.hdfc.bank.in/ Beneficiary Name: Test User Account Number: 1234567890 IFSC Code: HDFC0001234 Amount: 5000	Funds should be transferred successfully	Funds transferred successfully with confirmation message	Passed
Req_1.5	TC_06	Verify user can view past transactions	1.User should open Microsoft Edge browser 2. User should navigate to https://www.hdfc.bank.in/ 3. User should login to hdfc bank application 4.User should navigate to "Transaction History" section 5.User should select account number 6.User should choose date range for transactions 7.User should click on "Search" or "View" option 8.User should review displayed transaction records 9.User should verify accuracy of transaction details	Url: https://www.hdfc.bank.in/ Date Range: Last 30 days	Transaction history should be displayed correctly	Transaction details displayed accurately	Passed

Req_1.6	TC_07	Verify user can add, edit, and delete beneficiaries	1.User should open Microsoft Edge browser 2. User should navigate to https://www.hdfc.bank.in/ 3. User should login to hdfc bank application 4.User should navigate to "Beneficiary Management" section 5.User should click on "Add Beneficiary" option 6.User should enter beneficiary details 7.User should confirm addition using OTP 8.User should verify beneficiary added successfully 9.User should edit beneficiary details and save changes 10.User should delete beneficiary entry 11.User should confirm deletion	Url: https://www.hdfc.bank.in/ Beneficiary Name: Test User Account Number: 1234567890 IFSC Code: HDFC0005678	Beneficiary should be managed successfully	Beneficiary added, edited, and deleted successfully	Passed
Req_1.7	TC_08	Verify user can update personal details	1.User should open Microsoft Edge browser 2. User should navigate to https://www.hdfc.bank.in/ 3. User should login to hdfc bank application with valid credentials 4.User should navigate to "Profile" section 5.User should select "Edit Profile" option 6.User should update personal details (email, phone number) 7.User should save updated information 8.User should verify confirmation message 9.User should check updated details in profile	Url: https://www.hdfc.bank.in/ Email: asish@test.com Mobile: 9876543210	Profile details should be updated successfully	Profile details updated successfully	Passed
Req_1.8	TC_09	Verify user can logout securely	1.User should open Microsoft Edge browser 2. User should navigate to https://www.hdfc.bank.in/ 3. User should login to hdfc bank application with valid credentials 4.User should access dashboard page 5.User should click on "Logout" option 6.User should confirm logout action if prompted 7.User should be redirected to login page 8.User should verify session is terminated 9.User should attempt to access dashboard using back button	Url: https://www.hdfc.bank.in/ Email: asish@test.com Password: *****	User should be logged out and redirected to login page	User logged out successfully and redirected to login page	Passed

Defect Report

Defect ID	Defect Description	Steps to Reproduce	Priority	Severity	Status	Created By	Assigned To	Screenshot
1	System allows login with weak password without enforcing strong password policy validation.	1. User should open Microsoft Edge browser. 2. User should navigate to https://www.hdfcbank.com/ 3. User should click on NetBanking Login option. 4. User should enter valid Customer ID. 5. User should attempt to set/reset password using weak password (e.g., "123456" or "password"). 6. User should observe password validation response.	High	Critical	New	Asish Kumar Khandual	Raghavendra BN	 1 Case Study 1 System allows login with weak password without enforcing strong password policy validation. HDFC BANK Welcome to HDFC Bank NetBanking Login to NetBanking Customer ID 12345678 Password ***** Forgot Customer ID / Password? LOGIN Now to NetBanking? Register Now © Copyright HDFC Bank Ltd. Privacy Policy Terms and Conditions Security User entered Customer ID and a weak password ("123456") and was able to login successfully without any password policy validation or error.
2	System allows multiple registrations using the same mobile number without duplication validation.	1. User should open browser and navigate to http://www.google.com/ 2. User should click on "Click to follow the link." 3. User should enter personal details including mobile number already registered. 4. User should proceed with OTP verification. 5. User should observe system acceptance.	High	Major	New	Asish Kumar Khandual	Raghavendra BN	 2 Case Study 2 System allows multiple registrations using the same mobile number without duplication validation. HDFC BANK Open Savings Account 1 Enter Details 2 Verify OTP 3 Set Login 4 Complete Personal Details Full Name Asish Kumar Khandual Mobile Number 9876543210 Email ID asish.kumar@gmail.com By proceeding, you agree to the Terms & Conditions. CANCEL PROCEED Features - Zero Balance Account - Instant Account Opening - Digital Onboarding - Secure Banking User registered a new account using mobile number 9876543210 which was already used for a previous registration. System allowed registration without any duplicate mobile number validation.

3	User account is created or login session initiated without mandatory OTP verification.	1.User should navigate to HDFC NetBanking login page. 2.User should enter valid Customer ID. 3.User should choose OTP-based authentication. 4.User should skip entering OTP or manipulate request flow. 5.User should observe access granted.	High	Critical	New	Asish Kumar Khandual	Raghavendra BN	
4	Incorrect auto-population of branch/location details based on IFSC or PIN code input.	1.User should open HDFC website. 2.User should navigate to "Find Branch/ATM" or account opening form 3.User should enter valid PIN code or IFSC code. 4.User should observe auto-filled location details.	Medium	Minor	New	Asish Kumar Khandual	Raghavendra BN	
5	User is able to access dashboard and perform limited transactions despite incomplete KYC/profile details.	1.User should register for HDFC NetBanking. 2.User should login with valid credentials. 3.User should skip mandatory KYC/profile completion steps. 4.User should navigate to dashboard. 5.User should attempt basic operations (balance check, fund transfer).	Medium	Major	New	Asish Kumar Khandual	Raghavendra BN	

Requirement Traceability Matrix

Req No.	Test Case ID	Requirement Description	Status
R_1.0	TC_01	The System shall allow new user to successfully register on HDFC.BANK.COM platform using valid personal details including name, email ID, mobile number, password, and location, with mandatory OTP verification.	TC_01 – Passed
R_1.1	TC_02	The System shall allow registered user to log in to HDFC.BANK.COM application using valid credentials with secure authentication mechanisms.	TC_02 – Passed
R_1.2	TC_03	The System shall redirect authenticated user to HDFC.BANK.COM dashboard displaying account summary and navigation modules after successful login.	TC_03 – Passed
R_1.3	TC_04	The System shall allow user to view HDFC.BANK.COM accurate account balance details within dashboard/account summary section.	TC_04 – Passed
R_1.4	TC_05	The System shall enable user to transfer funds securely to HDFC.BANK.COM registered beneficiary using NEFT/IMPS/RTGS with OTP confirmation.	TC_05 – Passed
R_1.5	TC_06	The System shall allow user to view HDFC.BANK.COM transaction history with correct records based on selected account and date range.	TC_06 – Passed
R_1.6	TC_07	The System shall allow user to HDFC.BANK.COM add, edit, and delete beneficiaries securely with validation and OTP verification.	TC_07 – Passed
R_1.7	TC_08	The System shall allow user to HDFC.BANK.COM update profile details such as email ID and mobile number with proper validation and confirmation.	TC_08 – Passed
R_1.8	TC_09	The System shall allow user to securely log out from HDFC.BANK.COM application and terminate session, preventing unauthorized access.	TC_09 – Passed

Peer Review

ASISH <kumarkhandual@gmail.com>

To: Raghavendra Bn <raghu.astehead@gmail.com>

Sun, 21 Jun 2026 at 1:54 pm

Hi Raghavendra,

I have shared the test deliverables for peer review. Please find the Test Plan, Test Scenarios, Test Cases, Defect Report, and RTM attached.

Kindly review these documents and share your feedback or suggestions at your convenience.

Thank you for your time and support.

Best regards,
Asish Kumar Khandual



Raghavendra Bn

to me ▼

Sun 4 Jan, 19:06 (3 days ago)



Dear **Harshit Rajput**,

Hope this email finds you well...!

Thank you very much for promptly sharing the Test Plan and the Testing Files. I appreciate the time and effort you have invested in preparing these documents and ensuring that they are comprehensive and well-organized.

I will take some time to thoroughly review both the Test Plan and the Testing Files to ensure that all requirements and scenarios have been adequately covered. If I come across any areas that require clarification or if there are any suggestions for improvement, I will reach out to you with my feedback and any follow-up questions.

Your diligence and attention to detail are truly valued, and I am confident that your contributions will greatly assist in the smooth execution of our testing process. Please feel free to let me know if there is any additional information you would like to share or if you have any specific points you would like me to focus on during my review.

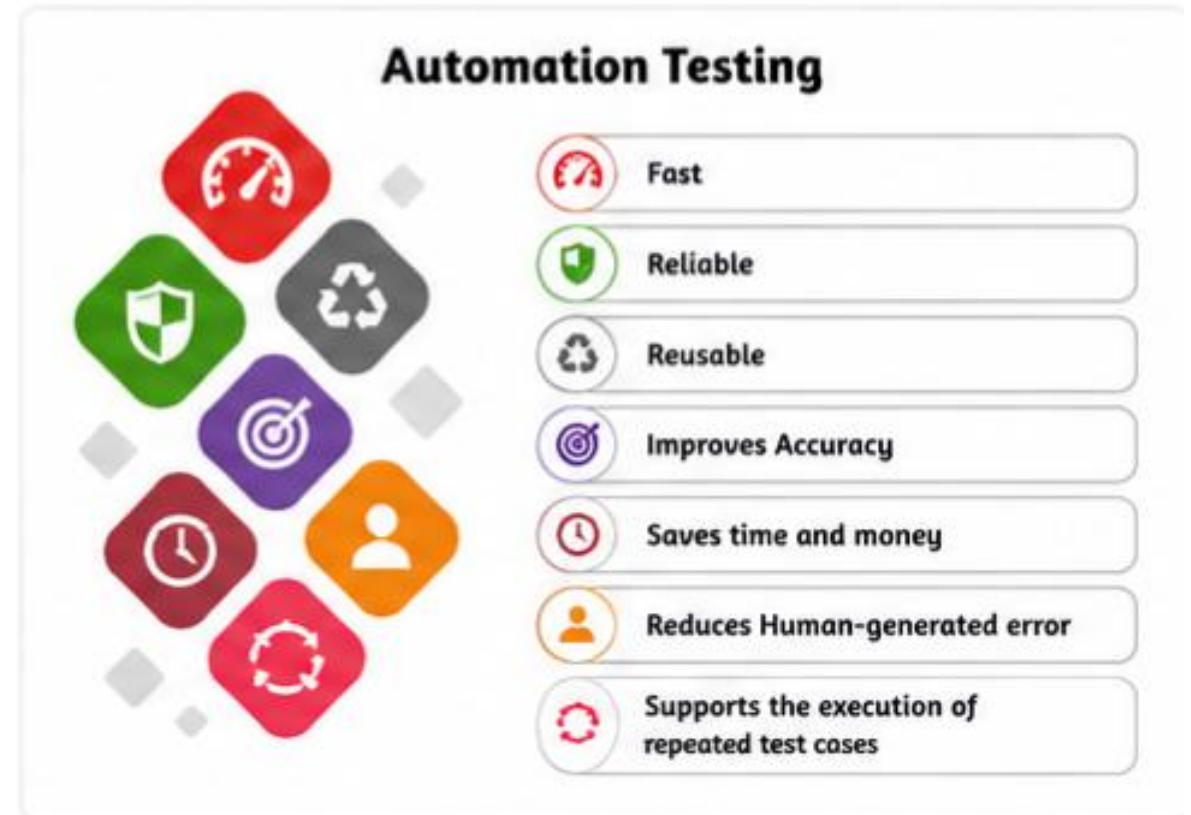
Thank you once again for your hard work and timely submission.

Best regards,
Raghavendra BN

Automation Testing

Automation testing is a software testing method where test cases are run automatically using tools and scripts to check whether the application works correctly.

Automation testing increases speed, accuracy, and reliability for repetitive and regression testing.



Tools & Setup

JDK(JAVA)	26(Latest Stable)
-----------	-------------------

Selenium	4.44.0
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TestNG	7.11.0
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Apache POI	5.5.1
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Postman	11.x
---------	------

Eclipse	2026
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Apache JMeter	5.6.3
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MySQL Workbench	8.0.43
-----------------	--------

Jenkins	LTS 2.516.1
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JDBC	
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Folder Structure

▼ > ParaBankAutomation [Selenium Java Program main]

- src/main/java
- src/main/resources
- ▼ > src/test/java
 - ▼ > com.Framework.Pages
 - > BaseClass.java
 - > LoginParaBank.java
 - ▼ > com.Framework.TestCases
 - > ExecuteLoginPage.java
 - ▼ > com.Utility
 - > BrowserFactory.java
 - > ConfigDataProvider.java
 - > ExcelDataProvider.java
 - > Helper.java
- src/test/resources
- > JRE System Library [JavaSE-1.8]
- > Maven Dependencies
- > > allure-results
- > > Configuration
- > > ScreenShots
- > > src
- > target
- > > TestData
- > > test-output
- pom.xml



▼ > src/test/java

- ▼ > com.Framework.Pages
 - > BaseClass.java
 - > LoginParaBank.java
- ▼ > com.Framework.TestCases
 - > ExecuteLoginPage.java
- ▼ > com.Utility
 - > BrowserFactory.java
 - > ConfigDataProvider.java
 - > ExcelDataProvider.java
 - > Helper.java



▼ Configuration

- config.properties

▼ ScreenShots

- ParaBank_06_20_2026_22_19_02.png
- ParaBank_06_20_2026_22_36_52.png
- ParaBank_06_20_2026_23_04_08.png
- ParaBank_06_20_2026_23_05_50.png

POM.xml

```
Bind to grammar/schema...
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="https://maven.apache.org/POM/4.0.0"
3   xmlns:xsi="https://www.w3.org/2001/XMLSchema-instance"
4   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
5   <modelVersion>4.0.0</modelVersion>
6   <groupId>HybridDrivenFramework</groupId>
7   <artifactId>HybridDrivenFramework</artifactId>
8   <version>0.0.1-SNAPSHOT</version>
9   <dependencies>
10    <!-- Source:
11     https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
12    <dependency>
13      <groupId>org.seleniumhq.selenium</groupId>
14      <artifactId>selenium-java</artifactId>
15      <version>4.44.0</version>
16      <scope>compile</scope>
17    </dependency>
18
19    <!-- Source: https://mvnrepository.com/artifact/org.testng/testng -->
20    <dependency>
21      <groupId>org.testng</groupId>
22      <artifactId>testng</artifactId>
23      <version>7.12.0</version>
24      <scope>test</scope>
25    </dependency>
26
27    <!-- Source: https://mvnrepository.com/artifact/au.com.bytecode.opencsv -->
28    <dependency>
29      <groupId>au.com.bytecode</groupId>
30      <artifactId>opencsv</artifactId>
31      <version>2.4</version>
32      <scope>compile</scope>
33    </dependency>
34
35    <!-- Source: https://mvnrepository.com/artifact/org.apache.poi/poi -->
36    <dependency>
37      <groupId>org.apache.poi</groupId>
38      <artifactId>poi</artifactId>
39      <version>5.5.1</version>
40      <scope>compile</scope>
41    </dependency>
42
43    <!-- Source: https://mvnrepository.com/artifact/org.apache.poi/poi-ooxml -->
44    <dependency>
45      <groupId>org.apache.poi</groupId>
46      <artifactId>poi-ooxml</artifactId>
47      <version>5.5.1</version>
48      <scope>compile</scope>
49    </dependency>
50
51    <!-- Source:
52     https://mvnrepository.com/artifact/io.qameta.allure/allure-testng -->
53    <dependency>
54      <groupId>io.qameta.allure</groupId>
```

```
55      <artifactId>allure-testng</artifactId>
56      <version>2.35.2</version>
57      <scope>compile</scope>
58    </dependency>
59
60    <!-- Source:
61     https://mvnrepository.com/artifact/org.apache.maven.plugins/maven-surefire-report-plugin -->
62    <dependency>
63      <groupId>org.apache.maven.plugins</groupId>
64      <artifactId>maven-surefire-report-plugin</artifactId>
65      <version>3.5.6</version>
66      <scope>compile</scope>
67    </dependency>
68
69    <!-- Source:
70     https://mvnrepository.com/artifact/com.relevantcodes/extentreports -->
71    <dependency>
72      <groupId>com.relevantcodes</groupId>
73      <artifactId>extentreports</artifactId>
74      <version>2.41.2</version>
75      <scope>compile</scope>
76    </dependency>
77
78    <dependency>
79      <groupId>org.apache.logging.log4j</groupId>
80      <artifactId>log4j-core</artifactId>
81      <version>2.23.1</version>
82    </dependency>
83
84  </dependencies>
85 </project>
```


Helper.java

```
package com.Utility;

import java.io.File;

public class Helper {

    public static void captureScreenshot(WebDriver driver) {

        try {
            TakesScreenshot ts = (TakesScreenshot) driver;
            File source = ts.getScreenshotAs(OutputType.FILE);
            String filePath = "./ScreenShots/ParaBank_" + getCurrentDateTime() + ".png";

            FileUtils.copyFile(source, new File("./ScreenShots/ParaBank_" + getCurrentDateTime() + ".png"));
            System.out.println("Captured ScreenShot - On Failure");

        } catch (Exception e) {
            System.out.println("Exception while taking screenshot" + e.getMessage());
        }

    }

    public static String getCurrentDateTime() {

        DateFormat customFormat = new SimpleDateFormat("MM_dd_yyyy_HH_mm_ss");
        Date currentDate = new Date();
        return customFormat.format(currentDate);

    }

}
```

BrowserFactory.java

```
package com.Utility;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.edge.EdgeDriver;
import org.openqa.selenium.firefox.FirefoxDriver;

public class BrowserFactory {

    static WebDriver driver;

    public static WebDriver BrowserOptions(WebDriver driver, String browserName, String AppUrl) {

        if (browserName.equalsIgnoreCase("chrome")) {
            driver = new ChromeDriver();
        }

        else if (browserName.equalsIgnoreCase("edge")) {
            driver = new EdgeDriver();
        }

        else if (browserName.equalsIgnoreCase("firefox")) {
            driver = new FirefoxDriver();
        }

        else {
            System.out.println("Browser Does Not Support....");
        }

        driver.manage().window().maximize();
        driver.get(AppUrl);
        return driver;

    }

    public static void quitBrowser(WebDriver driver) {
        driver.quit();
    }

}
```

ConfigDataProvider.java

```
1 package com.Utility;
2
3 + import java.io.File;
4
5
6
7 public class ConfigDataProvider {
8
9     Properties pro;
10
11 - public ConfigDataProvider() {
12     File src = new File("./Configuration/config.properties");
13
14 -     try {
15         FileInputStream fis = new FileInputStream(src);
16         pro = new Properties();
17         pro.load(fis);
18     }
19
20 -     catch (Exception e) {
21         System.out.println("Unable to load excel file " + e.getMessage());
22     }
23 }
24
25 - public String getBrowser() {
26     return pro.getProperty("Browser");
27 }
28
29 - public String getAppUrl() {
30     return pro.getProperty("AppUrl");
31 }
32
33 }
```

ExcelDataProvider.java

```
1 package com.Utility;
2
3 + import java.io.File;
4
5
6
7 public class ExcelDataProvider {
8
9     XSSFWorkbook wb;
10
11 - public ExcelDataProvider() {
12     File src = new File("./TestData/testdata.xlsx");
13
14 -     try {
15         FileInputStream fis = new FileInputStream(src);
16         wb = new XSSFWorkbook(fis);
17     }
18
19 -     catch (Exception e) {
20         System.out.println("Unable to load Xls file " + e.getMessage());
21     }
22 }
23
24 - public String getStringData(int sheetIndex, int row, int column) {
25     return wb.getSheetAt(sheetIndex).getRow(row).getCell(column).getStringCellValue();
26 }
27
28 - public String getStringData(String sheetName, int row, int column) {
29     return wb.getSheet(sheetName).getRow(row).getCell(column).getStringCellValue();
30 }
31
32 - public double getNumericData(String sheetName, int row, int column) {
33     return wb.getSheet(sheetName).getRow(row).getCell(column).getNumericCellValue();
34 }
35
36 }
37
```

BaseClass.java

```
1 package com.Framework.Pages;
2
3 import org.openqa.selenium.WebDriver;
4
5 public class BaseClass {
6
7     public WebDriver driver;
8     public ExcelDataProvider excel;
9     public ConfigDataProvider config;
10
11     @BeforeSuite
12     public void SetUp() {
13         excel = new ExcelDataProvider();
14         config = new ConfigDataProvider();
15     }
16
17     @BeforeClass
18     public void BrowserTest() {
19         driver = BrowserFactory.BrowserOptions(driver, config.getBrowser(), config.getAppUrl());
20     }
21
22     @AfterClass
23     public void tearDown() {
24         BrowserFactory.quitBrowser(driver);
25     }
26
27     @AfterMethod
28     public void tearDownMethod(ITestResult result) {
29         if(result.getStatus() == ITestResult.FAILURE) {
30             Helper.captureScreenshot(driver);
31         }
32     }
33 }
```

LoginPage.java

```
1 package com.Framework.Pages;
2
3 import org.openqa.selenium.WebDriver;
4
5 public class LoginParaBank {
6     WebDriver driver;
7
8     public LoginParaBank(WebDriver driver) {
9         this.driver = driver;
10     }
11
12     @FindBy(xpath = "//input[@name='username']")
13     WebElement email;
14
15     @FindBy(xpath = "//input[@name='password']")
16     WebElement password;
17
18     @FindBy(xpath = "//input[@value='Log In']")
19     WebElement loginbtn;
20
21     public void login_ParaBank(String uname, String pass) {
22         email.sendKeys(uname);
23         password.sendKeys(pass);
24         loginbtn.click();
25     }
26 }
```

BaseClass.java

```
1 package com.Framework.TestCases;
2
3+ import org.openqa.selenium.support.PageFactory;
9
  Run ALL
10 public class ExecutelLoginPage extends BaseClass {
11     @Test
12     Run / Debug
12- public void loginApp() throws Exception {
13
14     ExcelDataProvider excel = new ExcelDataProvider();
15     excel.getStringData("Sheet1", 0, 0);
16
17     LoginParaBank loginTestData = PageFactory.
18         initElements(driver, LoginParaBank.class);
19
20
21     loginTestData
22         .login_ParaBank(excel.getStringData("Sheet1", 0, 0),
23         excel.getStringData("Sheet1", 0, 1));
24     Thread.sleep(5000);
25
26 }
27 }
```



```
27
28- public String getStringData(String sheetName, int row, int column) {
29     return wb.getSheet(sheetName).getRow(row).getCell(column).getStringCellValue();
30 }
```



```
public void login_ParaBank(String uname, String pass) {
    email.sendKeys(uname);
    password.sendKeys(pass);
    loginbtn.click();
}
```


Allure Report



Allure



Overview



Categories



Suites



Behaviors



Packages



Graphs



Timeline



Dark theme



Language

Allure Report 6/22/2026

11:37:43 - 11:37:52 (9s 435ms)

1

test case

100%

Suites 1 item total

Default suite

1

Show all

Environment

There are no environment variables

Trend

There is nothing to show

Categories 0 items total

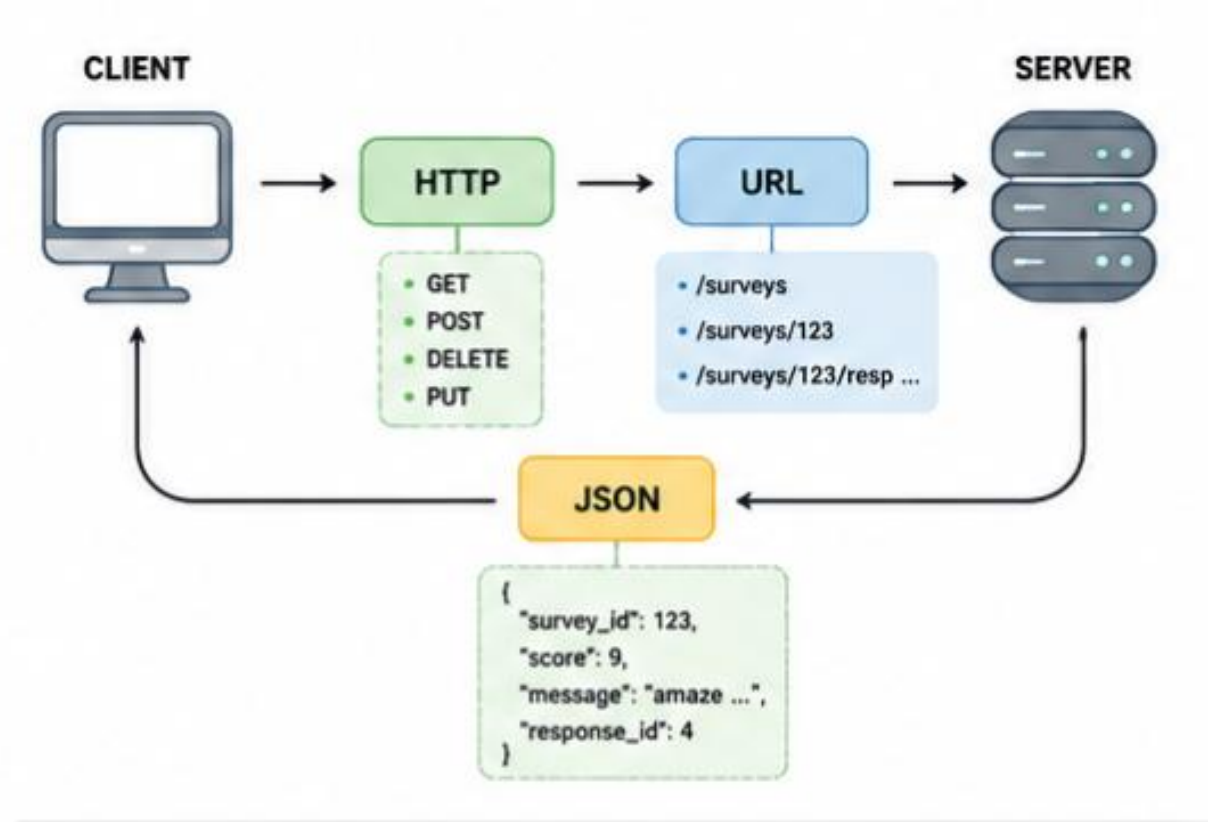
Show all

Executors

API Testing

API testing is the process of testing application programming interfaces to ensure they function correctly, securely, and efficiently.

It focuses on validating the communication between different software components by checking requests, responses, status codes, and data accuracy without relying on the user interface.



Postman

The screenshot displays the Postman application interface. On the left sidebar, under 'COLLECTIONS', the 'GET Request' folder is expanded, showing a collection named 'GET api.open-meteo.com'. The main workspace shows a GET request to 'https://api.open-meteo.com/v1/forecast?latitude=28.61&longitude=77.23¤t_weather=true'. The response is a 200 OK status with a JSON body. The bottom status bar indicates the response time is 1.25 s and the body size is 658 B.

BootCampTesting2026 > GET Request > api.open-meteo.com

GET https://api.open-meteo.com/v1/forecast?latitude=28.61&longitude=77.23¤t_weather=true

Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Document this request... Write with AI

Body Cookies Headers 5 Test Results 200 OK • 1.25 s • 658 B • Save Response

```
{
  "latitude": 28.576448,
  "longitude": 77.18678,
  "generationtime_ms": 0.16558170318603516,
  "utc_offset_seconds": 0,
  "timezone": "GMT",
  "timezone_abbreviation": "GMT",
  "current_weather": {
    "weather": "partlycloudy",
    "temperature": 28.5,
    "windspeed": 1.2,
    "humidity": 65,
    "cloudcover": 35
  }
}
```

Connect Git Console Terminal 0 0 Cloud Agent Globals Vault Tools

Home

BootCampTesting

Search

Invite

Upgrade

Notifications

Settings

Profile

Overview

Overview

BootCampTe

GET api.open-me

POST restfulapi.n

PUT jsonplaceho

POST jsonplaceh

+

No environment

⋮

HTTP

BootCampTesting2026 > POST Request > restfulapi.net

Save

Share

POST

https://restfulapi.net/

Send

Docs

Params

Authorization

Headers 9

Body

Scripts

Settings

Cookies

☐ none

☐ form-data

☐ x-www-form-urlencoded

☒ raw

☐ binary

☐ GraphQL

JSON

Schema

Beautify

```
1 {
2   "id": 123,
3   "title": "What is REST",
4   "content": "REST is an architectural style for building web services...",
5   "published_at": "2023-11-04T14:30:00Z",
6   "author": {
7     "id": 456,
```

Body

Cookies

Headers 17

Test Results

200 OK

794 ms

141.73 KB

Save Response

⋮

HTML

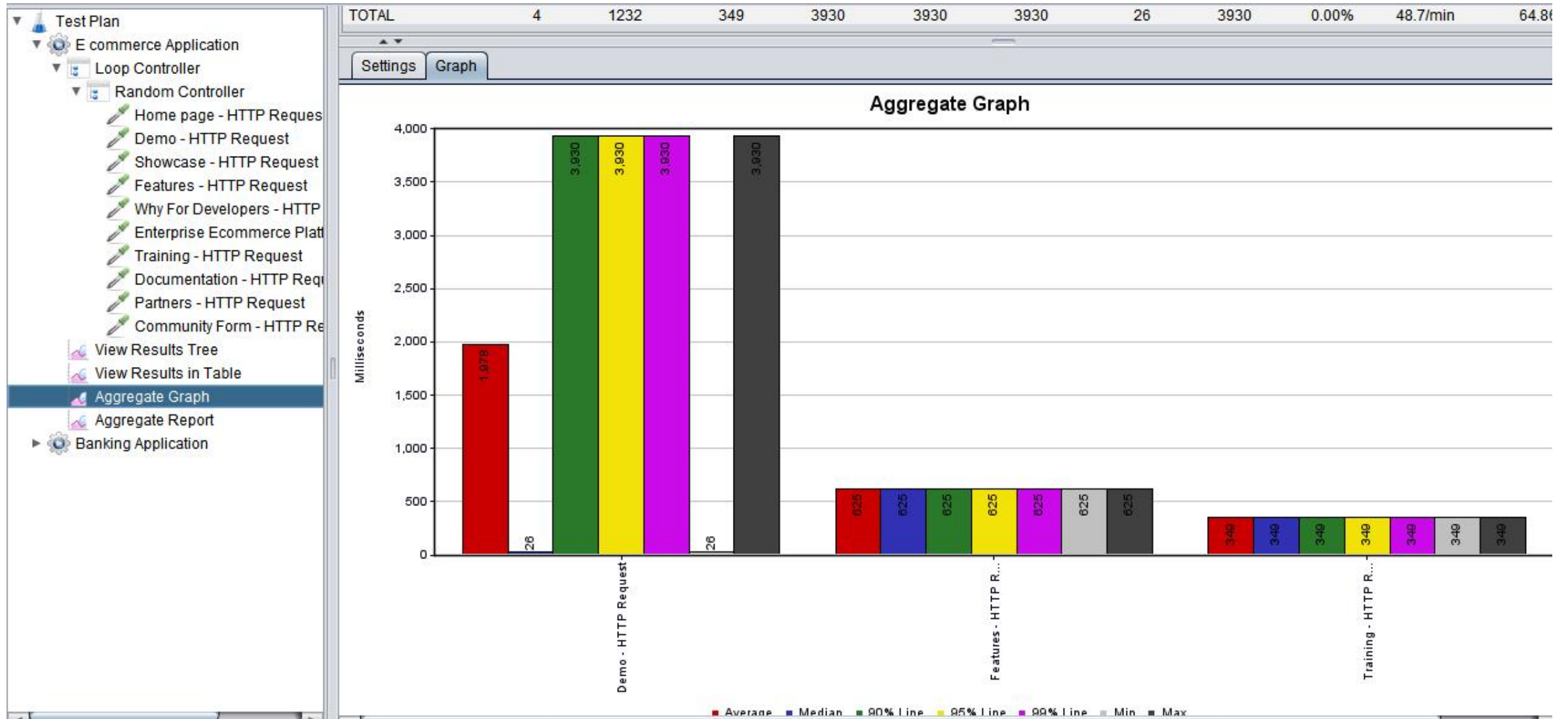
Preview

Visualize

⋮

```
1 <!DOCTYPE html>
2 <html lang="en-US">
3 <head>
4   <meta charset="UTF-8">
5   <meta name='robots' content='index, follow, max-image-preview:large, max-snippet:-1, max-video-preview:-1' />
6 <meta name="viewport" content="width=device-width, initial-scale=1">
7   <title>What is REST?: REST API Tutorial</title>
```


JMeter Report



Apache JMeter Dashboard

[Dashboard](#)

[Charts](#)

[Customs Graphs](#)

Test and Report information

Source file	"testdata.csv"
Start Time	"6/22/26, 9:45 AM"
End Time	"6/22/26, 9:45 AM"
Filter for display	""

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.625	500 ms	1 sec 500 ms	Total
0.000	500 ms	1 sec 500 ms	Investment - HTTP Request
0.500	500 ms	1 sec 500 ms	Features - HTTP Request
0.500	500 ms	1 sec 500 ms	Checking - HTTP Request
0.750	500 ms	1 sec 500 ms	Business leading - HTTP Request
0.750	500 ms	1 sec 500 ms	Training - HTTP Request
1.000	500 ms	1 sec 500 ms	Partners - HTTP Request

Requests Summary



Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label ^	#Samples ♦	FAIL ♦	Error % ♦	Average ♦	Min ♦	Max ♦	Median ♦	90th pct ♦	95th pct ♦	99th pct ♦	Transactions/s ♦	Received ♦	Sent ♦
Total	8	0	0.00%	735.63	77	1505	691.50	1505.00	1505.00	1505.00	2.14	74.18	0.29
Business leading - HTTP Request	2	0	0.00%	793.50	455	1132	793.50	1132.00	1132.00	1132.00	1.26	9.98	0.19
Checking - HTTP Request	1	0	0.00%	648.00	648	648	648.00	648.00	648.00	648.00	1.54	12.54	0.23
Features - HTTP Request	1	0	0.00%	1003.00	1003	1003	1003.00	1003.00	1003.00	1003.00	1.00	141.31	0.12
Investment - HTTP Request	1	0	0.00%	1505.00	1505	1505	1505.00	1505.00	1505.00	1505.00	0.66	5.53	0.10
Partners - HTTP Request	1	0	0.00%	330.00	330	330	330.00	330.00	330.00	330.00	3.03	115.20	0.36
Training - HTTP Request	2	0	0.00%	406.00	77	735	406.00	735.00	735.00	735.00	2.45	80.13	0.29

DataDriveTesting.java

```
1 package com.DataDriverTestingCSVFile;
2
3 + import java.io.FileReader;[]
13
14 public class DataDriverTestingCSV {
15     WebDriver driver;
16     String CsvPath = "D:\\Workbook1.csv";
17
18     @BeforeTest
19 - public void SetUp() {
20         driver = new ChromeDriver();
21         driver.navigate().to("https://only-testing-blog.blogspot.com/2014/05/form.html");
22         driver.manage().window().maximize();
23     }
24
25     @Test
26 - public void CSVFile() throws Exception {
27         CSVReader reader = new CSVReader(new FileReader(CsvPath));
28         String[] csvCell;
29
30 -         while ((csvCell = reader.readNext()) != null) {
31             String FName = csvCell[0];
32             String LName = csvCell[1];
33             String Email = csvCell[2];
34             String MNumb = csvCell[3];
35             String CName = csvCell[4];
36
37             driver.findElement(By.name("FirstName")).sendKeys(FName);
38             driver.findElement(By.name("LastName")).sendKeys(LName);
39             driver.findElement(By.name("EmailID")).sendKeys(Email);
40             driver.findElement(By.name("MobNo")).sendKeys(MNumb);
41             driver.findElement(By.name("Company")).sendKeys(CName);
42
43             Thread.sleep(8000);
44             driver.findElement(By.xpath("//input[@value='Submit']")).click();
45             Thread.sleep(8000);
46             driver.switchTo().alert().accept();
47         }
48     }
49
50     @AfterTest
51 - public void endTest() {
52         driver.quit();
53     }
54 }
```


Cucumber Testing

The screenshot shows an IDE with three tabs: 'Cucumber1st...', 'LoginTest.f...', and 'CucumberTes...'. The 'Cucumber1st...' tab is active, showing a Gherkin test scenario:

```
1 Feature: Checking Login functionality  Run | Debug | Profile
2
3 Scenario: Successful Login with valid Credentials  Run | Debug | Profile
4   Given User should Open Chrome Browser
5   When User should Enter Url in Browser
6   When User should Navigate Home Page
7   And Enter UserName and Password in edit box
8   And Click On Login PushButton
9   Then Message displayed Login Successfully
10
11
12
```

The 'Console' tab is also open, showing the following output:

```
<terminated> LoginTest.feature [Cucumber-JVM] C:\Program Files\openjdk-21.0.2\jdk-21
Jun 22, 2026 11:28:39 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find an exact match for CDP version 149, returning the closest version; found:
Test Login | Practice Test Automation
Login Title: Logged In Successfully | Practice Test Automation

1 scenarios (1 passed)
6 steps (6 passed)
0m 10.422s
```

The Project Explorer on the left shows the following structure:

- ✓ M>J > Cucumber1stTest [Cucumber Automation Testing]
 - ✓ > src/test/java
 - ✓ > StepDefinition
 - > > CucumberTestCase.java
 - > JRE System Library [jdk-21.0.2]
 - > Maven Dependencies
 - ✓ > Features
 - > LoginTest.feature
 - > TestCase2.feature
 - > TestCase3.feature
 - > > src
 - > > target
 - > > pom.xml

Rest Assured

The screenshot displays the Eclipse IDE interface with the following components:

- Package Explorer:** Shows the project structure for 'rest.assured', including 'src/main/java', 'src/main/resources', 'src/test/java', 'src/test/resources', 'JRE System Library [JavaSE-1.8]', 'Maven Dependencies', 'src', 'target', 'test-output', 'pom.xml', and 'testng.xml'.
- Code Editor:** Displays the file 'CompletePutPostTest.java'. The code includes package declarations, imports, and a test class with a static final String 'BASE_URL', a JSON object 'testUser', an integer 'createdUserId', and a '@BeforeClass' annotated 'setup()' method that initializes these variables.
- Problems View:** Shows the 'Results of running class CompletePutPostTest' with a summary: 'Passed: 5', 'Failed: 0', 'Skipped: 0', and 'ests: 1/1 Methods: 5 (6393 ms)'.
- Test Results View:** Lists the test methods and their execution times:
 - testPostRequestComplete (2.278 s)
 - testPutRequestComplete (0.724 s)
 - testPostRequestBDDComplete (1.068 s)
 - testPutRequestBDDComplete (0.9 s)
 - testPutAndPostWithExtraction (0.668 s)
- Console View:** Shows the output of the test run, including the request body and response details for a POST request.

Console Output:

```
<terminated> CompletePutPostTest [TestNG] C:\Users\Bappi.Gorain\p2\pool\plugins\org.e
===== POST REQUEST - TRADITIONAL =====
Request Body: {
  "name": "Test User",
  "id": 1,
  "email": "raghu.asteapahead@gmail.com",
  "username": "raghavendra"
}

--- POST Response Details ---
1. Status Code: 201
2. Complete Response: {
  "name": "Test User",
  "id": 11,
  "email": "raghu.asteapahead@gmail.com",
  "username": "raghavendra"
}
```

The Windows taskbar at the bottom shows the system clock at 08:35 PM on 22-06-2026, along with weather (30°C Partly cloudy), search, and various system icons.

Git Repositories

Module Name:-	GitHub Repositories	Submitted By:-	Asish Kumar khandual
No of Repositories :-	18	Submitted To:-	Raghavendra BN
Emp ID:- 85009841			
SL No.	Date	Project Name	Repository Link
1	27-06-2026	5 Bug in Any Application Quality of Software Software-Development-Life-Cycle-model Boundary Value Analysis (BVA) Equivalence Class Partitioning (ECP) Decision tree Test Plan , Test Scenario , Test Case , Bug Report , RTM Report	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/27-06-Manual_Testing
2	30-06-2026	JAVA-TRAINING-CODE	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/30-06-Java_Training
3	01-07-2026	Abstract-class-casestudy, Exception-Handling-CaseStudy Inheritance-CaseStudy,DataDrivenTestingUsingPOI DataDrivenTestingUsingCSV	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/01-07-Java
4	02-07-2026	DataProvideInTestNG, DifferentBrowserTestCaseStudy	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/02-07-Selenium_1.0
5	03-07-2026	NavigationCommandWebDriver, HybridDrivenFrameWork	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/03-07-Selenium_1.1
6	04-07-2026	ParameterTestNG, PageObjectModelEasyCalculation	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/04-07-Selenium_1.2
7	05-07-2026	SeleniumWebDriverLocator-CaseStudy, SeleniumWebDriverLocator	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/05-07-SeleniumWebDrivers_1.0
8	06-07-2026	SeleniumWebDriverAction, SeleniumWebDiverDropDownBox	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/06-07-SeleniumWebDrivers_1.1
9	08-07-2026	SeleniumWebDriverAlerts, SeleniumWebDriverMultiFrame	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/08-07-SeleniumWebDrivers_1.2
10	09-07-2026	SeleniumWebDriverScreenShot,SeleniumWebDriverRadioButton	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/09-07-SeleniumWebDrivers_1.3
11	10-07-2026	TestNGAssertion, SynchronizationSelenium	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/10-07-TestNG
12	11-07-2026	TestNGDifferentBrowser, JMeter-CaseStudy1	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/11-07-JMeter
13	12-07-2026	JMeter-CaseStudy2, JMeterPracticeExample	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/12-07-JMeter_1.0
14	13-07-2026	ChasePage-CaseStudy, ProfileBrightest-CaseStudy, PracticeAutomation-CaseStudy	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/13-07-CaseStudy
15	14-07-2026	Selenium-Case-Study-1, Selenium-Case-Study-2, Selenium-Case-Study-6, Selenium-Case-Study-7, Selenium-Case-Study-8, Selenium-Case-Study-9 Selenium-Case-Study-10, Selenium-Case-Study-11, Selenium-Case-Study-12 Selenium-Case-Study-13, Selenium-Case-Study-14, Selenium-Java-Automation	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/14-07-SeleniumCaseStudy
16	14-07-2026	JMeter Case Study	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/14-07-JMeterCaseStudy
17	17-07-2026	Cucumber Case Study	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/17-07-CucumberAutomation
18	19-07-2026	POSTMANCaseStudy	https://github.com/Asishkhandual/BootCampTesting2026/tree/main/19-07-PostMan



THANK YOU

for your time and attention

